

# Over-the-Air Decentralized Federated Learning Under MIMO Noisy Channel

Zhiyuan Zhai, Xiaojun Yuan, *Senior Member, IEEE*, and Xin Wang, *Fellow, IEEE*

**Abstract**—Decentralized federated learning (DFL) is an emerging paradigm for leveraging the rapidly growing data from wireless devices in a fully distributed manner. However, the deployment of DFL is facing some pivotal challenges, including communication bottlenecks due to extensive inter-device message exchanges and the difficulty for edge devices to achieve consensus. To address these challenges, this paper proposes to employ the over-the-air computation (Aircomp) technique to improve communication efficiency and introduces a mixing matrix mechanism to guarantee consensus. Specifically, we present a novel multiple-input multiple-output over-the-air DFL (MIMO OA-DFL) framework for addressing the DFL design problem in general *ad hoc* networks. A rigorous convergence bound is derived to quantitatively capture the impact of mixing matrix and communication error on the system performance. The results show that the communication errors, the spectral gap of the mixing matrix, and the mixing matrix itself have a significant impact on the learning performance. Building on this result, we formulate a joint communication-learning optimization problem to optimize transceiver beamformers and mixing matrix. Numerical experiments demonstrate the substantial performance enhancement achieved by our proposed scheme.

**Index Terms**—Decentralized federated learning, over-the-air model aggregation, consensus problem.

## I. INTRODUCTION

Empowered by an unprecedented increase in local data generated by mobile edge devices, there is a surging trend in developing deep learning applications at the edge of wireless networks. Federated learning (FL), as a distributed machine learning paradigm, is an emerging solution well-suited to meet this demand [1]. However, the deployment of FL faces significant limitations due to its high reliance on the central PS, such as communication congestion and reduced fault tolerance. These limitations become particularly challenging when dealing with a large number of edge devices. Moreover, in scenarios like autonomous robotics and collaborative driving [2], FL may not be a viable solution due to the absence of an available central PS. To address these drawbacks, decentralized federated learning (DFL) has been introduced as a promising alternative. In DFL, each edge device uses device-to-device (D2D) communication to exchange learned models. This distributed mechanism eliminates the need for the central PS, thereby reducing network congestion and enhancing system robustness.

In light of these potentials, DFL has become a prominent research area while many existing works, e.g., [3], [4], explore DFL from the network layer. However, these works are based on the assumption of error-free communication links but the real-world communication systems often suffer from imperfections. Limited wireless resources, and channel distortion

caused by deep fading, thermal noise as well as inter-device interference compromise the accuracy of model exchanges and hence the DFL learning performance [5]. Based on this consideration, several recent works [6], [7], [8] begin to focus on the communication aspect of DFL and propose over-the-air computation (Aircomp) [9] to improve the communication efficiency. Over-the-air computation is a technique using the superposition property of electromagnetic waves to enable devices to transmit their models using shared radio resources. Specifically, [6] uses a heuristic greedy coloring algorithm to determine the communication order and enable devices to perform over-the-air computation sequentially. Similarly, [7] separates the communication process into scheduling and transmission parts and appoints a certain device as the central server to produce interference-free over-the-air aggregation. The authors in [8] propose a one-step over-the-air scheme where all devices exchange models in a single phase to improve the training speed.

Nevertheless, all these recent works have their weaknesses. Particularly, [6] and [7] determine the mixing matrix, i.e., the matrix collects the contribution of each D2D link, based on standard examples, which may not be suitable for dynamic wireless DFL environment. Moreover, the heuristic protocols employed in their designs do not guarantee the optimality of the DFL performance. Although [8] has evidenced the effectiveness of the over-the-air technique in improving DFL model transmission performance, this work only focuses on communication beamforming optimization in a fully connected topology. That is, the lack of considerations from learning perspectives prevents it from fully realizing the potential of DFL. Therefore, there is a pressing need to address DFL problems in general network topologies from a joint communication-learning perspective.

In this paper, we present a novel multiple-input multiple-output over-the-air decentralized federated learning (MIMO OA-DFL) framework to address the aforementioned issues. We conduct rigorous convergence analysis to characterize the impact of mixing matrix and communication errors on the DFL learning accuracy under moderate assumptions. Our analysis provides a quantitative insight into what factors exactly constrain the performance when deploying DFL in wireless environment. Based on this, we propose a communication-learning co-design algorithm that utilizes alternating optimization (AO) to jointly optimize the mixing matrix and transceiver beamformers. Simulation results demonstrate the effectiveness of the proposed scheme and shed light on the variance in performance across different topologies within the MIMO OA-

DFL framework.

*Notations:* We use  $[M]$  to denote the set  $\{i | 1 \leq i \leq M\}$ , and denote the real and complex number sets by  $\mathbb{R}$  and  $\mathbb{C}$ , respectively;  $(\cdot)^*$ ,  $(\cdot)^T$ ,  $(\cdot)^H$ , and  $(\cdot)^\dagger$  denote the conjugate, the transpose, the conjugate transpose, and the pseudoinverse, respectively;  $\mathcal{CN}(\mu, \sigma^2)$  denotes circularly-symmetric complex normal distribution with mean  $\mu$  and covariance  $\sigma^2$ . The  $l_2$ -norm is denoted by  $\|\cdot\|$ , while the Frobenius norm is denoted by  $\|\cdot\|_F$ . Besides, we use  $\mathbf{1}_n$  to denote the column vector in  $\mathbb{R}^n$  with all elements being 1, and  $\mathbf{1}$  to denote such a vector with the appropriate dimension;  $\overline{(\cdot)}$  denotes the mean of a variable,  $\oslash$  denotes the Hadamard division operator, and  $\lambda_i(\cdot)$  denotes the  $i$ -th largest eigenvalue of a matrix.

## II. LEARNING AND COMMUNICATION MODELS

### A. Decentralized Federated Learning

In DFL system,  $M$  devices cooperatively train a machine learning model, and the common objective of the devices is to minimize an empirical loss function

$$f(\mathbf{x}) = \frac{1}{M} \sum_{i=1}^M f_i(\mathbf{x}), \quad (1)$$

where  $\mathbf{x} \in \mathbb{R}^D$  is the model parameter with dimension  $D$ , and  $f_i: \mathbb{R}^D \rightarrow \mathbb{R}$  is the local loss function of device  $i$  defined by

$$f_i(\mathbf{x}) := \mathbb{E}_{\xi_i \sim \mathcal{D}_i} F(\mathbf{x}, \xi_i), \quad (2)$$

with  $\mathcal{D}_i$  being the predefined distribution of local data samples on device  $i$ , and  $F(\mathbf{x}, \xi_i)$  being the loss function with respect to samples  $\xi_i$ .

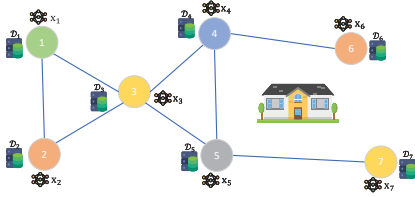


Fig. 1. An example of the DFL system with seven devices.

To describe the communication topology, we use  $e_{ij}$  as an indicator function of the communication link between device  $i$  and device  $j$ . That is,  $e_{ij} = 1$  if the communication link between device  $i$  and device  $j$  exists, and  $e_{ij} = 0$  otherwise. We assume full-duplex communication, i.e.,  $e_{ij} = e_{ji}$ . We say that device  $i$  is a neighbor of device  $j$  if  $e_{ij} = 1$ . An example of DFL system is shown in Fig. 1.

We adopt the stochastic gradient descent method for local training, where the model parameters of all devices are iteratively updated at each training round. At the  $t$ -th round, the DFL training process consists of the following three steps:

- *Local gradient computation:* Each device  $i$  computes the local gradient  $\nabla F(\mathbf{x}_i^{(t)}, \xi_i^{(t)})$  where  $\mathbf{x}_i^{(t)}$  (or  $\xi_i^{(t)}$ ) denotes the model parameter (or sample) of device  $i$  in round  $t$ .

- *Gossip model aggregation:* Each device fetches the model parameters from its neighbors through wireless channels and estimates the weighted average as

$$\mathbf{x}_i^{(t+\frac{1}{2})} = \sum_{j=1}^M w_{ij}^{(t)} \mathbf{x}_j^{(t)}, \quad \forall i \in [M] \quad (3)$$

where  $w_{ij}^{(t)} \in [0, 1]$  is the weighting factor for device  $j$  aggregating on device  $i$  at round  $t$ . Note that  $w_{ij}^{(t)} = 0$  if device  $i$  does not have a communication link with device  $j$ . We refer to  $\mathbf{x}_i^{(t+\frac{1}{2})}$  as the ideal (error-free) aggregation model at device  $i$ , and denote by  $\hat{\mathbf{x}}_i^{(t+\frac{1}{2})}$  an estimate of  $\mathbf{x}_i^{(t+\frac{1}{2})}$ . Due to the communication noise and channel fading, we generally have  $\hat{\mathbf{x}}_i^{(t+\frac{1}{2})} \neq \mathbf{x}_i^{(t+\frac{1}{2})}$ ,  $\forall i \in [M]$ .

- *Local model update:* Based on the estimate  $\hat{\mathbf{x}}_i^{(t+\frac{1}{2})}$ ,  $\forall i \in [M]$ , each device updates the local model parameter as

$$\mathbf{x}_i^{(t+1)} = \hat{\mathbf{x}}_i^{(t+\frac{1}{2})} - \lambda \nabla F(\mathbf{x}_i^{(t)}, \xi_i^{(t)}), \quad \forall i \in [M], \quad (4)$$

where  $\lambda \in \mathbb{R}$  represents the learning rate.<sup>1</sup>

For the sake of convenience, we use a mixing matrix, denoted by  $\mathbf{W}^{(t)} \in \mathbb{R}^{M \times M}$ , to represent the weighting factor of all devices at round  $t$ , with  $w_{ij}^{(t)}$  being the  $(i, j)$ -th element of  $\mathbf{W}^{(t)}$ .

### B. MIMO IBFD Communication Channel

We assume that each device is equipped with  $N_T$  transmit antennas and  $N_R$  receive antennas for full-duplex communication, yielding a multiple-input multiple-output (MIMO) in-band full-duplex (IBFD) *ad hoc* network. We further assume that the residual self-interference can be suppressed by using the self-interference cancellation (SIC) technique [10].

We consider a block-fading channel, where the channel keeps invariant within each communication round. The received signal of device  $i$  at the  $l$ -th channel use is given by

$$\mathbf{y}_i^{(t)}[l] = \sum_{j \in \mathcal{M}_i} \mathbf{H}_{\langle i, j \rangle}^{(t)} \mathbf{s}_j^{(t)}[l] + \mathbf{n}_i^{(t)}[l], \quad \forall i \in [M], \quad (5)$$

where  $\mathcal{M}_i$  denotes the neighbor set of device  $i$ ,  $\mathbf{H}_{\langle i, j \rangle}^{(t)} \in \mathbb{C}^{N_R \times N_T}$  denotes the channel matrix between the  $i$ -th device and the  $j$ -th device,  $\mathbf{s}_j^{(t)}[l] \in \mathbb{C}^{N_T}$  denotes the transmit signal of user  $j$  in the  $l$ -th channel use, and  $\mathbf{n}_i^{(t)}[l] \in \mathbb{C}^{N_R}$  is an additive white Gaussian noise (AWGN) vector with each element following the distribution  $\mathcal{CN}(0, \sigma_n^2)$ . Let  $L$  be the number of channels uses in each communication round. Then, the received signal matrix of each device can be expressed as

$$\mathbf{Y}_i^{(t)} = \sum_{j \in \mathcal{M}_i} \mathbf{H}_{\langle i, j \rangle}^{(t)} \mathbf{S}_j^{(t)} + \mathbf{N}_i^{(t)}, \quad \forall i \in [M], \quad (6)$$

<sup>1</sup>Note that the local model update and gossip model aggregation step can be exchanged. That is, the devices first update the local models and then average the updated models with neighbors, which does not hurt our theoretical analysis. When gossip model aggregation step is executed before local model update step, these two steps can be run in parallel.

where  $\mathbf{Y}_i^{(t)} \triangleq [\mathbf{y}_i^{(t)}[1], \dots, \mathbf{y}_i^{(t)}[L]]$ ,  $\mathbf{S}_j^{(t)} \triangleq [\mathbf{s}_j^{(t)}[1], \dots, \mathbf{s}_j^{(t)}[L]]$  and  $\mathbf{N}_i^{(t)} \triangleq [\mathbf{n}_i^{(t)}[1], \dots, \mathbf{n}_i^{(t)}[L]]$ .

### III. PROPOSED MIMO OA-DFL FRAMEWORK

In this section, we illustrate the proposed MIMO OA-DFL framework. Specifically, in each training round, each device computes the local gradient and then performs gossip model aggregation over the channel given in (6) based on over-the-air computation. After that, each device updates its local model according to (4).

In over-the-air aggregation, the following procedure is concurrently executed on every device. We first normalize the model parameter  $\mathbf{x}_i^{(t)}$  as

$$\tilde{\mathbf{x}}_i^{(t)} = (\mathbf{x}_i^{(t)} - \bar{x}_i^{(t)} \mathbf{1}_D) / \sqrt{v_i^{(t)}}, \quad \forall i \in [M] \quad (7)$$

where  $\bar{x}_i^{(t)} = \frac{1}{D} \sum_{d=1}^D x_i^{(t)}[d]$  and  $v_i^{(t)} = \frac{1}{D} \sum_{d=1}^D (x_i^{(t)}[d] - \bar{x}_i^{(t)})^2$  are the mean and variance of  $\mathbf{x}_i^{(t)}$ , respectively. Following the common practice [11], the mean and variance are exchanged between the neighbors via error-free links. Then, we convert  $\tilde{\mathbf{x}}_i^{(t)} \in \mathbb{R}^D$  to a complex version  $\mathbf{r}_i^{(t)} \in \mathbb{C}^L$

$$\mathbf{r}_i^{(t)} = \tilde{\mathbf{x}}_i^{(t)} \left(1 : \frac{D}{2}\right) + j\tilde{\mathbf{x}}_i^{(t)} \left(\frac{D+2}{2} : D\right), \quad \forall i \in [M], \quad (8)$$

where we choose the block length  $L = D/2$  for simplicity. Let  $\mathbf{u}_i^{(t)} \in \mathbb{C}^{N_T}$  be the multicast beamforming vector of device  $i$ . The transmit signal of the  $i$ -th device, denoted by  $\mathbf{S}_i^{(t)}$ , is

$$\mathbf{S}_i^{(t)} \triangleq \mathbf{u}_i^{(t)} (\mathbf{r}_i^{(t)})^T \in \mathbb{C}^{N_T \times L}, \quad (9)$$

and the corresponding power constraint is  $\mathbb{E} \|\mathbf{S}_i^{(t)}[l]\|^2 = 2 \|\mathbf{u}_i^{(t)}\|^2 \leq P_0, \forall i \in [M]$ , where  $P_0$  denotes the maximum transmit power for each device and  $\mathbf{S}_i^{(t)}[l] \triangleq \mathbf{r}_i^{(t)}[l] \mathbf{u}_i^{(t)} \in \mathbb{C}^{N_T}$  is the transmit signal of device  $i$  at the  $l$ -th channel use. Then the received signal of each device can be expressed as

$$\hat{\mathbf{r}}_i^{(t)} = \left( \sum_{j \in \mathcal{M}_i} \mathbf{r}_j^{(t)} (\mathbf{H}_{\langle i, j \rangle}^{(t)} \mathbf{u}_j^{(t)})^T + \mathbf{N}_{k, i}^T \right) (\mathbf{f}_i^{(t)})^*, \quad \forall i \in [M] \quad (10)$$

where  $\mathbf{f}_i^{(t)} \in \mathbb{C}^{N_R}$  represents the receive beamformer of device  $i$  used to retrieve the desired signal. Each device computes the estimate of  $\mathbf{x}_i^{(t+\frac{1}{2})}$  from  $\hat{\mathbf{r}}_i^{(t)}$  by

$$\tilde{\mathbf{x}}_i^{(t+\frac{1}{2})} = [\text{Re}\{\hat{\mathbf{r}}_i^{(t)}\}^T, \text{Im}\{\hat{\mathbf{r}}_i^{(t)}\}^T]^T \quad (11)$$

$$+ \tilde{x}_i^{(t)} \mathbf{1}_D + w_{i, i} \mathbf{x}_i^{(t)}, \quad \forall i \in [M] \quad (12)$$

where  $\tilde{x}_i^{(t)} \triangleq \sum_{j \in \mathcal{M}_i} w_{ij}^{(t)} \bar{x}_j^{(t)}$ . Note that the term  $\tilde{x}_i^{(t)} \mathbf{1}_D$  is added back to compensate the mean of  $\mathbf{x}_i^{(t)}$  subtracted in the step (7). Finally, with the received signal  $\tilde{\mathbf{x}}_i^{(t+\frac{1}{2})}$ , each device updates the local model based on (4).

### IV. CONVERGENCE ANALYSIS

In this section, we analyze the convergence of MIMO OA-DFL and reveal the impact of the mixing matrix  $\mathbf{W}^{(t)}$  and the beamformers  $\mathbf{f}^{(t)} \triangleq \{\mathbf{f}_i^{(t)}\}_{i=1}^M$  and  $\mathbf{u}^{(t)} \triangleq \{\mathbf{u}_i^{(t)}\}_{i=1}^M$  on the learning performance.

#### A. Assumptions

To begin with, we make the following assumptions.

**Assumption 1:** (Invariant Gossip matrix). We assume  $\mathbf{W} = \mathbf{W}^{(t)}, \forall t$ .<sup>2</sup> We assume  $\mathbf{W}$  is a symmetric doubly stochastic matrix, i.e.,  $\mathbf{W}^T = \mathbf{W}$ ,  $\mathbf{W}\mathbf{1} = \mathbf{1}$ ,  $\mathbf{1}^T \mathbf{W} = \mathbf{1}^T$  and  $\mathbf{W} \in [0, 1]^{M \times M}$ . We further assume  $\delta(\mathbf{W}) \triangleq (\max\{\|\lambda_2(\mathbf{W})\|, |\lambda_M(\mathbf{W})|\})^2 < 1$ .

**Assumption 2:** ( $\omega$ -smoothness). The functions  $f_1, \dots, f_M$  are all differentiable and the corresponding gradients  $\nabla f_1(\cdot), \dots, \nabla f_M(\cdot)$  are Lipschitz continuous with parameter  $\omega$ .

**Assumption 3:** (Bounded variance). The variance of the stochastic gradient  $\mathbb{E} \|\nabla F(\mathbf{x}, \xi_i) - \nabla f_i(\mathbf{x})\|^2$  and  $\mathbb{E} \|\nabla f_i(\mathbf{x}) - \nabla f(\mathbf{x})\|^2$  are bounded, i.e.,

$$\mathbb{E}_{\xi_i \sim \mathcal{D}_i} \|\nabla F(\mathbf{x}, \xi_i) - \nabla f_i(\mathbf{x})\|^2 \leq \alpha^2, \quad \forall \mathbf{x} \in \mathbb{R}^D, i \in [M],$$

$$\mathbb{E}_{i \sim [M]} \|\nabla f_i(\mathbf{x}) - \nabla f(\mathbf{x})\|^2 \leq \beta^2, \quad \forall \mathbf{x} \in \mathbb{R}^D.$$

Assumptions 1-3 are commonly used in the literature on decentralized stochastic optimization and gossip algorithm; see, e.g., [12], [4]. Assumption 1 is related to the mixing matrix. It is known that such a mixing matrix exists for every connected graph [12]. Assumption 3 ensures a bounded gap between the gradient of the sample-dependent loss  $\nabla F(\mathbf{x}, \xi_i)$ , and that of the overall loss  $\nabla f(\mathbf{x})$ .

#### B. Convergence Analysis of MIMO OA-DFL

For the proposed MIMO OA-DFL, we can establish the following convergence result.<sup>3</sup>

**Proposition 1:** Under Assumption 1-3, with  $\lambda \leq 1/\omega$ , we have

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left\| \nabla f \left( \frac{\mathbf{X}^{(t)} \mathbf{1}}{M} \right) \right\|^2 &\leq \frac{1}{\left(\frac{1}{2} - 27M\lambda^2 G(\mathbf{W})\right)} \\ &\left( \frac{f(\frac{\mathbf{X}^{(0)} \mathbf{1}}{M}) - f^*}{\lambda T} + \frac{\alpha^2}{M} + (3M\alpha^2\lambda^2 + 27M\beta^2\lambda^2)G(\mathbf{W}) \right. \\ &\left. + \frac{9G(\mathbf{W})}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\mathbf{E}^{(t)}\|_F^2 + \frac{1}{\lambda^2 M^2 T} \sum_{t=0}^{T-1} \mathbb{E} \|\mathbf{E}^{(t)} \mathbf{1}\|^2 \right) \end{aligned} \quad (13)$$

where the expectation is over the randomness of channel noise and stochastic data sampling,  $G(\mathbf{W}) \triangleq \frac{\omega^2}{(1 - \sqrt{\delta(\mathbf{W})})^2 - 27M\lambda^2\omega^2}$ ,  $\mathbf{X}^{(t)} \triangleq [\mathbf{x}_1^{(t)}, \dots, \mathbf{x}_M^{(t)}]$ ,  $\hat{\mathbf{X}}^{(t+\frac{1}{2})} \triangleq [\hat{\mathbf{x}}_1^{(t+\frac{1}{2})}, \dots, \hat{\mathbf{x}}_M^{(t+\frac{1}{2})}]$ ,  $\mathbf{E}^{(t)} \triangleq \mathbf{X}^{(t)} \mathbf{W} - \hat{\mathbf{X}}^{(t+\frac{1}{2})}$  and  $f^*$  denotes the minimum value of  $f(\mathbf{x})$ .

<sup>2</sup>To obtain a tractable convergence bound, we assume that the mixing matrix keeps invariant throughout the learning process. In the following system optimization, we use the obtained convergence bound as a designing guidance to optimize the mixing matrix in each round, as discussed in Section V-A.

<sup>3</sup>Proof of all propositions is omitted due to space limitation and can be found in the extended version of this paper [13].

Note that  $f(G(\mathbf{W}))$  is monotonically increasing with respect to  $G(\mathbf{W})$ . Since  $0 \leq \delta(\mathbf{W}) < 1$ ,  $G(\mathbf{W})$  is also monotonically increasing with respect to  $\delta(\mathbf{W})$ . Therefore, it can be readily concluded that the convergence bound (13) monotonically increases with respect to  $\delta(\mathbf{W})$ .

To derive tractable expression for  $\mathbb{E} \|\mathbf{E}^{(t)}\|_F^2$  and  $\mathbb{E} \|\mathbf{E}^{(t)} \mathbf{1}\|^2$ , we assume that the model parameters  $\{\tilde{\mathbf{x}}_i^{(t)} | i \in [M]\}$  are independent and the model parameter elements  $\{\tilde{x}_i^{(t)}[d] | d \in [D], \forall i \in [M]\}$  are independent and identically distributed. i.e.,

$$\mathbb{E} [\tilde{\mathbf{x}}_i^{(t)} (\tilde{\mathbf{x}}_j^{(t)})^T] = \mathbf{0}, \forall i \neq j, \mathbb{E} [\tilde{\mathbf{x}}_i^{(t)} (\tilde{\mathbf{x}}_i^{(t)})^T] = \mathbf{I}, \forall i. \quad (14)$$

We then establish that

**Proposition 2:** For the MIMO OA-DFL scheme, with the assumption in (14),  $\mathbb{E} \|\mathbf{E}^{(t)}\|_F^2$  and  $\mathbb{E} \|\mathbf{E}^{(t)} \mathbf{1}\|^2$  are given by

$$\begin{aligned} \mathbb{E} \|\mathbf{E}^{(t)}\|_F^2 &= C \sum_{p=1}^M \left( \sum_{i \in \mathcal{M}_p} 2 (w_{ip} v_p^{(t)})^2 - \right. \\ &\quad \left. 4 \sum_{i \in \mathcal{M}_p} w_{ip} \operatorname{Re} \left\{ v_p^{(t)} g_{\langle i,p \rangle}^{(t)} \right\} + 2 \sum_{i \in \mathcal{M}_p} g_{\langle i,p \rangle}^H g_{\langle i,p \rangle} + \sigma_n^2 \|\mathbf{f}_i^{(t)}\|^2 \right) \\ \mathbb{E} \|\mathbf{E}^{(t)} \mathbf{1}\|^2 &= \frac{CM^2}{n^2} \left( \sum_{p=1}^M \sum_{j \in \mathcal{M}_p} 2 (w_{ip} w_{jp} (v_p^{(t)})^2) - 4 \sum_{p=1}^M \sum_{j \in \mathcal{M}_p} \right. \\ &\quad \left. w_{ip} \operatorname{Re} \left\{ v_p^{(t)} g_{\langle j,p \rangle}^{(t)} \right\} + 2 \sum_{p=1}^M \sum_{j \in \mathcal{M}_p} g_{\langle j,p \rangle}^{(t)} g_{\langle i,p \rangle}^{(t)} + \sum_{i=1}^M \sigma_n^2 \|\mathbf{f}_i^{(t)}\|^2 \right) \end{aligned}$$

where  $g_{\langle i,p \rangle}^{(t)} = (\mathbf{f}_i^{(t)})^H \mathbf{u}_p^{(t)} \mathbf{H}_{\langle i,p \rangle}^{(t)}$ .

From Propositions 1 and 2, we see that the mixing matrix  $\mathbf{W}$  and the beamformers  $\{\mathbf{u}^{(t)}, \mathbf{f}^{(t)}\}$  jointly have impact on the learning performance. In the following, we propose a systematic communication (i.e., beamformers) and learning (i.e., mixing matrix) co-design algorithm to improve the performance of the MIMO OA-DFL system.

## V. SYSTEM OPTIMIZATION

### A. Problem Formulation

We conduct the system optimization in a round-by-round fashion. For a given decentralized topology, we minimize the round-based convergence bound derived from (13) by jointly optimizing  $\{\mathbf{W}^{(t)}, \mathbf{u}^{(t)}, \mathbf{f}^{(t)}\}$ . We omit the superscript  $t$  in the sequel for brevity. The optimization problem is then cast as

$$(P1): \min_{\mathbf{W}, \mathbf{f}, \mathbf{u}} \frac{\left( Q + RG(\mathbf{W}) + 9G(\mathbf{W}) \mathbb{E} \|\mathbf{E}\|_F^2 + \frac{1}{\lambda^2 M^2} \mathbb{E} \|\mathbf{E} \mathbf{1}\|^2 \right)}{\left( \frac{1}{2} - 27M\lambda^2 G(\mathbf{W}) \right)} \quad (15a)$$

$$\text{s.t. } w_{ij} = 0, \forall j \in [M], \forall i \notin \mathcal{M}_j, \quad \mathbf{W}^T = \mathbf{W}, \mathbf{W} \mathbf{1} = \mathbf{1}, \mathbf{W} \in [0, 1]^{M \times M}, \quad (15b)$$

$$\|\mathbf{u}_i\|^2 \leq P_0/2, \forall i \in [M], \quad (15c)$$

where  $Q = \frac{f(\frac{\mathbf{X}(\mathbf{1})}{M}) - f^*}{\lambda T} + \frac{\alpha^2}{M}$ ,  $R = 3M\alpha^2\lambda^2 + 27M\beta^2\lambda^2$ , and  $G(\mathbf{W}) = \frac{\omega^2}{(1 - \sqrt{\delta(\mathbf{W})})^2 - 27M\lambda^2\omega^2}$ .

P1 is a non-convex problem. Different from the existing solutions [7], [8], where the transceiver beamformers can be optimized alternately, even with fixed  $\mathbf{f}$  and  $\mathbf{u}$ , P1 is still non-convex due to the coupling of  $\mathbf{W}$  and  $\delta(\mathbf{W})$ . However, we discover that by exploiting the monotonicity of  $\delta(\mathbf{W})$  and the structural information of doubly stochastic matrix, P1 can be efficiently solved in an AO manner, as detailed in what follows.

### B. Optimizing Beamformers for Given Mixing Matrix

We first optimize  $\mathbf{u}$  and  $\mathbf{f}$  for given  $\mathbf{W}$ . Dropping the irrelevant terms, we have the following problem

$$(P2): \min_{\mathbf{f}, \mathbf{u}} d(\mathbf{W}, \mathbf{f}, \mathbf{u}) \triangleq 9G(\mathbf{W}) \mathbb{E} \|\mathbf{E}\|_F^2 + \frac{1}{\lambda^2 M^2} \mathbb{E} \|\mathbf{E} \mathbf{1}\|^2 \quad (16)$$

s.t. (15c). (17)

1) *Optimizing  $\mathbf{u}$  for fixed  $\mathbf{f}$ :* For a fixed  $\mathbf{f}$ ,  $\mathbf{u}$  can be determined by solving the following problem:

$$(P3): \min_{\mathbf{u}} \sum_{p=1}^M \mathbf{u}_p^H \mathbf{M}_p \mathbf{u}_p - 2 \operatorname{Re} \left\{ \sum_{p=1}^M \mathbf{n}_p^H \mathbf{u}_p \right\} \quad (18)$$

$$\text{s.t. (15c).} \quad (19)$$

where

$$\begin{aligned} \mathbf{M}_p &= 9G(\mathbf{W}) \sum_{i \in \mathcal{M}_p} \mathbf{H}_{\langle i,p \rangle}^H \mathbf{f}_i \mathbf{f}_i^H \mathbf{H}_{\langle i,p \rangle} \\ &\quad + \frac{1}{\lambda^2} \frac{1}{M^2} \sum_{i,j \in \mathcal{M}_p} \mathbf{H}_{\langle i,p \rangle}^H \mathbf{f}_i \mathbf{f}_j^H \mathbf{H}_{\langle j,p \rangle}, \\ \mathbf{n}_p &= 9G(\mathbf{W}) \sum_{i \in \mathcal{M}_p} w_{ip} v_p (\mathbf{f}_i^H \mathbf{H}_{\langle i,p \rangle})^H \\ &\quad + \frac{1}{\lambda^2} \frac{1}{M^2} \sum_{i,j \in \mathcal{M}_p} v_p w_{ip} (\mathbf{f}_j^H \mathbf{H}_{\langle j,p \rangle})^H. \end{aligned}$$

Clearly,  $\mathbf{M}_p, \forall p \in [M]$  is a positive semidefinite matrix and hence P3 is a convex problem, which can be solved by off-the-shelf convex optimization tools, e.g., CVX [14].

2) *Optimizing  $\mathbf{f}_p$  for fixed  $\mathbf{u}$  and  $\{\mathbf{f}_i\}_{i \neq p}$ :* With fixed  $\mathbf{u}$  and  $\{\mathbf{f}_i\}_{i \neq p}$ , problem P2 reduces to

$$(P5): \min_{\mathbf{f}_p} \mathbf{f}_p^H \mathbf{A}_p \mathbf{f}_p - 4 \operatorname{Re} \{ \mathbf{b}_p^H \mathbf{f}_p \} \quad (21)$$

where

$$\begin{aligned} \mathbf{A}_p &= (18G(\mathbf{W}) + 2 \frac{1}{\lambda^2} \frac{1}{M^2}) \sum_{j \in \mathcal{M}_p} \mathbf{H}_{\langle p,j \rangle} \mathbf{u}_j \mathbf{u}_j^H \mathbf{H}_{\langle p,j \rangle}^H \\ &\quad + (\frac{1}{\lambda^2} \frac{1}{M^2} + 9G(\mathbf{W})) \sigma_n^2 \mathbf{I}_{N_R}, \\ \mathbf{b}_p &= 9G(\mathbf{W}) \sum_{j \in \mathcal{M}_p} w_{pj} v_j (\mathbf{u}_j^H \mathbf{H}_{\langle p,j \rangle})^H + \frac{1}{\lambda^2} \frac{1}{M^2} \sum_{i=j \in \mathcal{M}_i, \mathcal{M}_p} \sum_{i=j \in \mathcal{M}_i, \mathcal{M}_p} w_{ij} v_j \\ &\quad \left( \mathbf{u}_j^H \mathbf{H}_{\langle p,j \rangle} \right)^H - \frac{1}{\lambda^2} \frac{1}{M^2} \sum_{i=1, i \neq p}^n \sum_{j \in \mathcal{M}_p, \mathcal{M}_i} (\mathbf{f}_i^H \mathbf{H}_{\langle i,j \rangle} \mathbf{u}_j \mathbf{u}_j^H \mathbf{H}_{\langle p,j \rangle}^H)^H. \end{aligned}$$

This is an unconstrained convex problem, and the optimal solution is  $\mathbf{f}_p^* = 2\mathbf{A}_p^{-1} \mathbf{b}_p, \forall p \in [M]$ .



### C. Optimizing Mixing Matrix for Given Beamformers

What remains is to optimize the mixing matrix. For given  $\mathbf{u}$  and  $\mathbf{f}$ , P1 can be expressed as

$$(P5) : \min_{\mathbf{W}} \frac{Q + RG(\mathbf{W}) + 9G(\mathbf{W}) \mathbb{E} \|\mathbf{E}\|_F^2 + \frac{1}{\lambda^2 M^2} \mathbb{E} \|\mathbf{E}\mathbf{1}\|^2}{\frac{1}{2} - 27M\lambda^2 G(\mathbf{W})} \quad (23a)$$

$$\text{s.t. (15b)} \quad (23b)$$

We introduce slack variable  $\hat{\delta}$  and reformulate problem P5 as

$$(P6) : \min_{\mathbf{W}, \hat{\delta}} \frac{Q + RG(\hat{\delta}) + 9G(\hat{\delta}) \mathbb{E} \|\mathbf{E}\|_F^2 + \frac{1}{\lambda^2 M^2} \mathbb{E} \|\mathbf{E}\mathbf{1}\|^2}{\frac{1}{2} - 27M\lambda^2 G(\hat{\delta})} \quad (24a)$$

$$\text{s.t. } \delta(\mathbf{W}) \leq \hat{\delta}, \text{ (15b).} \quad (24b)$$

where  $G(\hat{\delta}) = \frac{\omega^2}{(1 - \sqrt{\hat{\delta}})^2 - 27M\lambda^2 \omega^2}$ .

**Proposition 3:** Problem P6 is equivalent to P5.

We can now optimize  $\mathbf{W}$  and  $\hat{\delta}$  in an alternating manner.

1) *Optimizing  $\mathbf{W}$  for fixed  $\hat{\delta}$ :* For a fixed  $\hat{\delta}$ , P6 reduces to

$$(P7) : \min_{\mathbf{W}} \hat{d}(\mathbf{W}, \mathbf{f}, \mathbf{u}) \text{ s.t. } \delta(\mathbf{W}) \leq \hat{\delta}, \text{ (15b).} \quad (25)$$

where  $\hat{d}(\mathbf{W}, \mathbf{f}, \mathbf{u}) = 9G(\hat{\delta}) \mathbb{E} \|\mathbf{E}\|_F^2 + \frac{1}{\lambda^2 M^2} \mathbb{E} \|\mathbf{E}\mathbf{1}\|^2$ .

Both  $\mathbb{E} \|\mathbf{E}^{(t)}\|_F^2$  and  $\mathbb{E} \|\mathbf{E}^{(t)}\mathbf{1}\|^2$  are convex quadratic functions of  $\mathbf{W}$ ; hence, the objective in P7 is convex. It can be shown that the sum of the two largest squared eigenvalues i.e.,  $\lambda_1(\mathbf{X}^2) + \lambda_2(\mathbf{X}^2)$  is convex with respect to symmetric matrix  $\mathbf{X}$ . For symmetric doubly stochastic matrix  $\mathbf{W}$ , we actually have  $\delta(\mathbf{W}) = \lambda_1(\mathbf{W}^2) + \lambda_2(\mathbf{W}^2) - 1$ . It then follows that the constraint  $\delta(\mathbf{W}) \leq \hat{\delta}$  is also convex. Hence, P7 is a convex problem that can be efficiently solved by e.g., CVX tools.

2) *Optimizing  $\hat{\delta}$  with fixed  $\mathbf{W}$ :* With fixed  $\mathbf{W}$ ,  $\hat{\delta}$  can be directly updated by  $\hat{\delta} = \delta(\mathbf{W})$  in each iteration.

## VI. SIMULATION RESULTS

In this section, we provide numerical results to demonstrate the effectiveness of the proposed MIMO OA-DFL scheme.

### A. Simulation Under Error Free Case

To start with, we consider an error-free case and perform DFL training with different mixing matrices. We conduct the learning task of image classification on the MNIST dataset [15]. We use 20k samples to train the model and 10k samples for validation from the original data set. The heterogeneous data splitting scheme in [16] is implemented, and we train a convolutional neural network (CNN) with two  $5 \times 5$  convolution layers (separately with 10 and 20 channels and each followed by  $2 \times 2$  max pooling), a subsequent batch normalization layer, a fully connected layer containing 50 units with ReLu activation and a final softmax output layer. The cross-entropy loss is used as the loss function.

In Figs. 2 and 3, we plot the minimum test accuracy (among all devices) and the average test accuracy (of the global model average) with different mixing matrices. We set the number of devices  $M = 30$ , learning rate = 0.02, momentum = 0.9, and the results are averaged over 30 Monte Carlo trials.

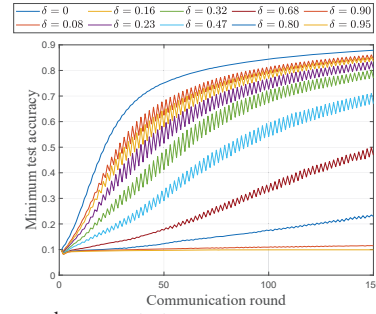


Fig. 2. Minimum and average test accuracy versus communication round for different choices of the mixing matrix.

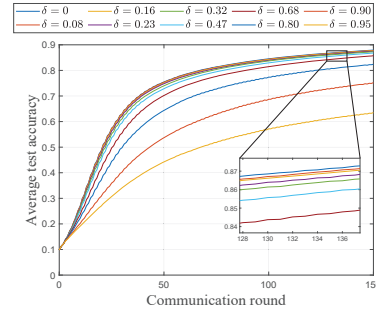


Fig. 3. Minimum and average test accuracy versus communication round for different choices of the mixing matrix.

As illustrated in Figs. 2 and 3, we observe that the test accuracy gradually deteriorates as the increase of  $\delta(\mathbf{W})$  in both subgraphs, which matches our analysis in Proposition 1 well. In the Fig. 2, we see that only the minimum accuracy of  $\delta(\mathbf{W}) = 0$  (fully connected) can keep close to the accuracy curve of the global model average. For  $\delta(\mathbf{W})$  more than 0.8, the worst-case learning performance in the left subgraph is prominently poor (less than 0.3). This is because for the system with high  $\delta(\mathbf{W})$ , there are significant discrepancies among the local models, resulting in extremely poor performance for some devices. Therefore, it is obvious that  $\delta(\mathbf{W})$  has an important impact on the consensus performance.

### B. Performance Comparison With Benchmarks

We next present a comparison between the proposed algorithm and state-of-the-art schemes.

We conduct simulation under a 30% sparsity topology, i.e. for this topology we have  $\frac{\text{number of 0 elements}}{M^2} = 30\%$ . The communication channels are modeled by independently and identically distributed (i.i.d.) Rayleigh fading, and the signal-to-noise ratio (SNR) at the transmitter side, defined as  $P_0/\sigma_n^2$ , is set equal as 5 dB for all devices. We set the number of training round  $T = 150$ , the number of devices  $M = 30$ , maximum transmit power  $P_0 = 1$  W,  $N_T = N_R = 10$ , optimization-related parameters  $J_{\max} = 20$ ,  $I_{1\max} = 50$ ,  $I_{2\max} = 50$ ,  $\lambda = 0.02$ , and  $\omega = 0.1$ . Besides, the learning configuration remains the same as described in Section VI-A.

The state-of-the-art schemes used for comparison are : joint optimization with separate over-the-air aggregation (JO with SOA), where the communication latency is  $M$ -times larger than that of the proposed algorithm; digital communication without mixing matrix optimization (DC w/o MMO) [17];

zero-forcing beamforming without mixing matrix optimization (ZFB w/o MMO) [8]; MMSE beamforming without mixing matrix optimization (MB w/o MMO) [18]; error free communication with optimized mixing matrix (Error free case).

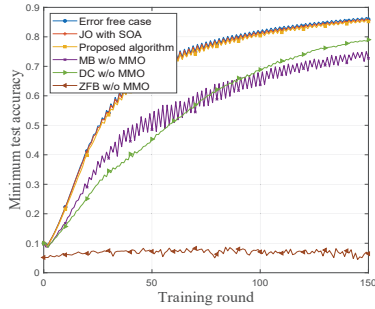


Fig. 4. Minimum test accuracy versus training round under 30% sparsity topology based on different schemes.

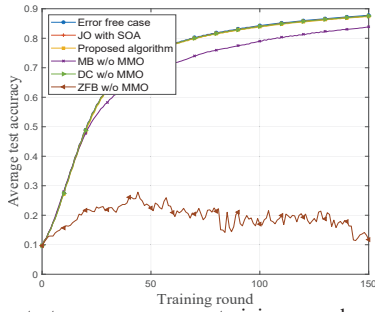


Fig. 5. Average test accuracy versus training round under 30% sparsity topology based on different schemes.

In Figs. 4 and 5, we compare the accuracy of the proposed algorithm with the benchmarks. The results demonstrate that the proposed algorithm achieves nearly the same accuracy as the JO with SOA scheme while consuming 30 times less communication time. Both of these schemes exhibit near-optimal performance as the error free case.

As for other benchmarks, we use the average point-wise accuracy loss (APL) i.e.,  $(\mathbf{a}_b - \mathbf{a}_p) \odot \mathbf{a}_p$  as a performance comparison metric, where  $\mathbf{a}_b \in \mathbb{R}^{150}$  (or  $\mathbf{a}_p \in \mathbb{R}^{150}$ ) is the learning accuracy of a certain benchmark (or proposed scheme) over 150 training rounds. We can see although DC w/o MMO scheme performs as well as proposed algorithm in average accuracy, it lags in minimum accuracy for 21.36% APL due to the limited consensus performance. Furthermore, the MB w/o MMO scheme suffers from both communication errors and discrepancies in local models, resulting in under-performance for 19.66% and 6.13% in minimum and average accuracy, respectively. Moreover, without considering channel noise, ZFB w/o MMO scheme cannot converge due to the huge communication error.

## VII. CONCLUSIONS

This paper developed the design of the MIMO OA-DFL system over wireless *ad hoc* networks. We used a mixing matrix mechanism to promote consensus and leveraged wireless beamforming to improve communication quality. Based on the derived convergence bound, we formulated a joint optimization problem with respect to transceiver beamformers and mixing matrix, and proposed a low-complexity algorithm to solve

this problem. Finally, simulation results demonstrated the communication and learning trade-off in different topologies and verified the superiority of our proposed algorithm to existing alternatives.

## REFERENCES

- [1] J. Konečný, H. B. McMahan, D. Ramage, and P. Richtárik, "Federated optimization: Distributed machine learning for on-device intelligence," *arXiv preprint arXiv:1610.02527*, 2016.
- [2] S. Savazzi, M. Nicoli, M. Bennis, S. Kianoush, and L. Barbieri, "Opportunities of federated learning in connected, cooperative, and automated industrial systems," *IEEE Commun. Mag.*, vol. 59, no. 2, pp. 16–21, 2021.
- [3] R. Olfati-Saber and R. M. Murray, "Consensus problems in networks of agents with switching topology and time-delays," *IEEE Trans. Automat. Contr.*, vol. 49, no. 9, pp. 1520–1533, 2004.
- [4] J. Wang and G. Joshi, "Cooperative SGD: A unified framework for the design and analysis of communication-efficient SGD algorithms," *arXiv preprint arXiv:1808.07576*, 2018.
- [5] H. Ye, L. Liang, and G. Y. Li, "Decentralized federated learning with unreliable communications," *IEEE J. Sel. Top. Signal Process.*, vol. 16, no. 3, pp. 487–500, 2022.
- [6] H. Xing, O. Simeone, and S. Bi, "Decentralized federated learning via SGD over wireless D2D networks," in *Proc. IEEE SPAWC*, 2020, pp. 1–5.
- [7] Y. Shi, Y. Zhou, and Y. Shi, "Over-the-air decentralized federated learning," in *Proc. IEEE ISIT*, 2021, pp. 455–460.
- [8] Z. Lin, Y. Gong, and K. Huang, "Distributed over-the-air computing for fast distributed optimization: Beamforming design and convergence analysis," *arXiv preprint arXiv:2204.06876*, 2022.
- [9] B. Nazer and M. Gastpar, "Computation over multiple-access channels," *IEEE Trans. Inf. Theory*, vol. 53, no. 10, pp. 3498–3516, 2007.
- [10] A. Sabharwal, P. Schniter, D. Guo, D. W. Bliss, S. Rangarajan, and R. Wichman, "In-band full-duplex wireless: Challenges and opportunities," *IEEE J. Sel. Areas Commun.*, vol. 32, no. 9, pp. 1637–1652, 2014.
- [11] H. Liu, X. Yuan, and Y.-J. A. Zhang, "Reconfigurable intelligent surface enabled federated learning: A unified communication-learning design approach," *IEEE Trans. Wireless Commun.*, vol. 20, no. 11, pp. 7595–7609, 2021.
- [12] A. Koloskova, S. Stich, and M. Jaggi, "Decentralized stochastic optimization and gossip algorithms with compressed communication," in *Proc. PMLR*, 2019, pp. 3478–3487.
- [13] Z. Zhai, X. Yuan, and X. Wang, "Decentralized federated learning via MIMO over-the-air computation: Consensus analysis and performance optimization," *arXiv preprint arXiv:2310.05075*, 2023.
- [14] S. Diamond and S. Boyd, "CVXPY: A Python-embedded modeling language for convex optimization," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2909–2913, 2016.
- [15] L. Deng, "The MNIST database of handwritten digit images for machine learning research [best of the web]," *IEEE Signal Process. Mag.*, vol. 29, no. 6, pp. 141–142, 2012.
- [16] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Artificial Intelligence and Statistics*. PMLR, 2017, pp. 1273–1282.
- [17] N. Yan, K. Wang, C. Pan, and K. K. Chai, "Performance analysis for channel-weighted federated learning in OMA wireless networks," *IEEE Signal Process. Lett.*, vol. 29, pp. 772–776, 2022.
- [18] C. Zhong, H. Yang, and X. Yuan, "Over-the-air federated multi-task learning over MIMO multiple access channels," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 3853–3868, 2023.